IoT Assignment 02-07-2026

Q1) You are building a smart irrigation system that must monitor soil moisture, control a water pump, and send data to a mobile application every 5 seconds. You have both an Arduino UNO and an ESP32. Which controller would you choose and why? Would your answer change if internet connectivity was not required?

Q2) A smart factory contains 500 sensors distributed across a 2 km area. Compare Wi-Fi, Bluetooth, LoRa, and MQTT-based communication. Which protocol or combination of protocols would you choose and why?

Q3) A temperature sensor suddenly starts showing impossible values (e.g., –120°C or 300°C). How you would determine whether the problem lies in the sensor, wiring, power supply, ADC, or software.

Q4) An ESP32-based weather station must operate on battery power for six months without maintenance. What hardware and software modifications would you make to maximize battery life?

Q5) An automatic braking system in a vehicle requires communication with almost zero delay. Which communication method would you choose and why?

Q6) You need to measure the average temperature of a classroom using only one sensor. Where would you place the sensor and why? Explain how poor placement could affect the accuracy of your measurements. Assume the classroom to be a square-base cuboid of side length “a” and height “b”.

Q7) Under what situations could an analog sensor perform better than a digital one?

Q8) Draw the complete architecture/block diagram of your IoT project, beginning from the sensor and ending at the user interface. Identify every possible point where a failure could occur and suggest one solution for each.

Q9) If an attacker gains access to your IoT device through the Wi-Fi network, what kinds of attacks could they perform? Suggest at least five methods to improve the security of the system (with applicable solutions, do not cite far-fetched ideas)

Q10) Your ESP32 must simultaneously interface with a DHT11 sensor, an ultrasonic sensor, an OLED display, a relay module, and a soil moisture sensor. What hardware or software conflicts could occur, and how would you solve them?

Q11) An ESP32 uploads sensor data to the cloud every minute. The internet connection fails for 12 hours. Design a strategy that ensures no important data is permanently lost.

Q12) Your IoT device must operate continuously for five years without human intervention. What changes would you make to both the hardware and software to improve long-term reliability?

Q13) Can an IoT system be considered "smart" if it does not use Artificial Intelligence? Justify your answer with suitable examples. What differentiates “smart” from “intelligent”?

Q14) Compare polling and interrupt-based programming in embedded systems. In which situations would interrupts significantly improve the performance of an IoT device?

Q15) Every engineering design involves trade-offs. Looking at your own IoT project, identify three design decisions where you sacrificed one aspect (such as cost, speed, power, accuracy, or complexity) to improve another. Explain whether you believe those trade-offs were justified.

Q16) Create a flowchart for an IoT system that stores sensor readings locally whenever Wi-Fi is unavailable and automatically uploads the stored data once the connection is restored. Mention all the programs/protocols/I-O used in the system.

Q17) Draw a flowchart showing how an ESP32 performs an OTA (Over-the-Air) firmware update, including error handling if the update fails.